

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - V

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Binary Search Tree: Write a program to:

- a. Create a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

print("root.right.left.data:", root.right.left.data)
print("\nDivyasnhu Singh So39")
```

Output:

cal 5.py =====:

root.right.left.data: E

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- b. Insert nodes into a binary search tree.

Code:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')
nodeH = TreeNode('H')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG
nodeF.right = nodeH

print("root.right.left.data:", root.right.right.left.data)
print("\nDivyasnhu Singh So39")
```

Output:

root.right.left.data: G

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- c. Search for a node in a binary search tree.

Code:

```
def search(root, key):
    if root is None:
        return False

    print("Checking:", root.data)

    if root.data == key:
        return True

    return search(root.left, key) or search(root.right, key)

key = input("Enter element to search: ")

# Search
if search(root, key):
    print("Element Found")
else:
    print("Element Not Found")
print("\nDivyasnhu Singh SO39")
```

Output:

```
root.right.left.data: G
Enter element to search: H
Checking: R
Checking: A
Checking: C
Checking: D
Checking: B
Checking: E
Checking: F
Checking: G
Checking: H
Element Found

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```